Detailed Report on

Adaptive Filter Design for CV QKD

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Abstract

The Continuous Variable (CV) Quantum Key Distribution (QKD) protocol facilitates the generation and secure distribution of symmetric cryptographic keys between two legitimate parties by leveraging standard telecommunication components. A critical aspect of CV-QKD implementation is the design of a robust Digital Signal Processing (DSP) framework capable of effectively mitigating various signal impairments, including offsets, noise, and hardware-induced distortions. In this work, we develop a dedicated DSP module tailored to operate reliably under a range of realistic vulnerabilities, ensuring the integrity and performance of the quantum signal throughout the key distribution process.The proposed DSP architecture focuses on three critical components: Root Raised Cosine (RRC) filtering, Frequency Domain Equalization (FDE), and in-phase/quadrature (I/Q) imbalance compensation. The implementation of RRC filtering effectively mitigates intersymbol interference (ISI), enabling bandwidth-efficient transmission and ensuring spectral matching between the transmitter and receiver. The FDE module successfully compensates for chromatic dispersion, maintaining signal integrity even over extended fiber links exceeding 100 km. To address I/Q imbalance, we employ a modified Gram–Schmidt Orthogonalization Procedure (GSOP), which preserves the quantum nature of the signal by avoiding amplitude normalization. The performance of the algorithm is rigorously evaluated under various challenging conditions, including high phase misalignments (up to 50°), significant gain imbalance, and low OSNR (15 dB). Simulation results demonstrate that the proposed method not only maintains a low bit error rate across these scenarios but also extends the maximum transmission distance from 40 km to 60 km, confirming its effectiveness and adaptability for long-distance, low-SNR quantum communication.

Introduction

This report details the derivation and implementation of pilot-assisted intradyne reception for high-speed continuous-variable quantum key distribution (CV-QKD) using a true local oscillator (LO), as well as frequency-domain chromatic dispersion (CD) compensation using overlap-save and overlap-add methods. To further improve time-domain performance and reduce inter-symbol interference (ISI), we also implement a Root Raised Cosine (RRC) pulse shaping and matched filtering scheme using a manually derived filter in the transmitter and receiver. we also compensate the I/Q imbalance of the signal. The combined simulation explores spectral behavior, ISI reduction, and potential SNR improvements through filter integration. The experimental and theoretical results are based on recent literature and simulation studies [1, 2].

1 Theoretical Background

1.1 Continuous-Variable Quantum Key Distribution (CV-QKD)

In CV-QKD, information is encoded in the quadrature amplitudes of weak coherent states:

$$|\alpha\rangle = |I_A + iQ_A\rangle$$

The quadrature measurement at the receiver is:

$$\Delta P \propto |\alpha_{LO}|(I\cos\Theta + Q\sin\Theta)$$

where Θ is the phase difference between the signal and the LO [1].

1.2 Chromatic Dispersion in Optical Fibers

The group velocity dispersion parameter β_2 and the chromatic dispersion parameter D are related by:

$$D = -\frac{2\pi c}{\lambda^2}\beta_2$$

where c is the speed of light and λ is the operating wavelength.

The total chromatic dispersion in single-mode fiber is:

$$D_{\text{total}} = D_{\text{material}} + D_{\text{waveguide}}$$

Material dispersion is given by:

$$D_{\text{material}} = -\frac{\lambda}{c}\frac{d^2n}{d\lambda^2}$$

where n is the refractive index [**?**].

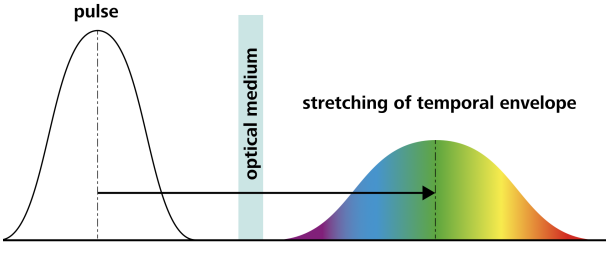


Figure 1: Visual Representation of Dispersion

1.3 Mathematical Foundation

The transfer function for chromatic dispersion compensation is given by:

$$H(\omega, z) = \exp\left(-j\frac{D\lambda^2 z\omega^2}{4\pi c}\right) \tag{1}$$

where:

- D = chromatic dispersion coefficient
- λ = operating wavelength
- z = fiber length
- c = speed of light
- ω = angular frequency

2 Convolution Theory and Digital Signal Processing

2.1 Linear Convolution Fundamentals

Linear convolution forms the mathematical foundation for digital signal processing in optical systems. For continuous-time signals, convolution is defined as:

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \tag{2}$$

For discrete-time signals, the linear convolution becomes:

$$y[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] \tag{3}$$

The convolution theorem in the frequency domain states:

$$\mathcal{F}\{y(t)\} = X(f) \cdot H(f) \tag{4}$$

where $\mathcal{F}\{\cdot\}$ denotes the Fourier transform.

2.2 Linear vs Circular Convolution

2.2.1 Linear Convolution Properties

For finite sequences of length N, linear convolution produces an output of length $2N-1$ samples. Consider two sequences:

- $x[n]$ with length N_x
- $h[n]$ with length N_h

The linear convolution $y_{\text{linear}}[n]$ has length $N_x + N_h - 1$.
Example Calculation:
For $x[n] = [1, 2, 3]$ and $h[n] = [4, 5, 6]$:

$$y[0] = x[0] \cdot h[0] = 1 \cdot 4 = 4 \tag{5}$$
$$y[1] = x[0] \cdot h[1] + x[1] \cdot h[0] = 1 \cdot 5 + 2 \cdot 4 = 13 \tag{6}$$
$$y[2] = x[0] \cdot h[2] + x[1] \cdot h[1] + x[2] \cdot h[0] = 1 \cdot 6 + 2 \cdot 5 + 3 \cdot 4 = 28 \tag{7}$$
$$y[3] = x[1] \cdot h[2] + x[2] \cdot h[1] = 2 \cdot 6 + 3 \cdot 5 = 27 \tag{8}$$
$$y[4] = x[2] \cdot h[2] = 3 \cdot 6 = 18 \tag{9}$$

Result: $y_{\text{linear}}[n] = [4, 13, 28, 27, 18]$.

2.2.2 Circular Convolution

Circular convolution is computed using FFT operations:

$$y_{\text{circular}}[n] = \text{IFFT}(\text{FFT}(x[n]) \cdot \text{FFT}(h[n])) \tag{10}$$

The circular convolution formula is:

$$y[n] = \sum_{k=0}^{N-1} x[k] \cdot h[(n-k) \bmod N] \tag{11}$$

This creates wraparound effects due to the modulo operation, resulting in time-domain aliasing.

2.3 Why Linear Convolution is Required

Linear convolution is essential in optical communication systems for several critical reasons:

1. **Causality Preservation:**

   Linear convolution maintains the causal relationship between input and output signals, which is essential for real-time processing. The output at any time instant depends only on current and past input values, ensuring physical realizability of the system.

2. **No Time-Domain Aliasing:**

   Unlike circular convolution, linear convolution doesn't create artificial wraparound effects that distort the signal. This prevents spurious frequency components from contaminating the desired signal spectrum.

3. **Filter Response Accuracy:**

   Linear convolution provides the true impulse response of the system, which is critical for chromatic dispersion compensation. The complete transient response is preserved, ensuring accurate equalization performance.

4. **Edge Effect Control:**

   Linear convolution properly handles signal boundaries without introducing spurious artifacts. This is particularly important for finite-length data blocks in practical implementations.

3 Frequency Domain Equalization Methods

3.1 Overlap-Save (OLS) Method

The overlap-save method divides the received signal into overlapping blocks for efficient frequency domain processing.

3.1.1 Algorithm Parameters

The fundamental parameters are related by:

$$L = N - R = N - M + 1 \tag{12}$$
$$R = M - 1 \tag{13}$$
$$N \geq M \tag{14}$$

where:

- N = FFT size (block length)
- L = number of valid output samples per block
- $R = M - 1$ = overlap size
- M = filter length

3.1.2 Block Formation

Each input block is formed as:

$$x_k[n] = x[kL - R + n], \quad n = 0, 1, \ldots, N - 1 \tag{15}$$

3.1.3 Processing Steps

The complete OLS algorithm follows these steps:

1. **Apply FFT:**

   Transform the input block to frequency domain:

   $$X_k[\omega] = \text{FFT}_N\{x_k[n]\} \tag{16}$$

2. **Multiply by Filter:**

   Apply the equalization filter in frequency domain:

   $$Y_k[\omega] = X_k[\omega] \cdot H[\omega] \tag{17}$$

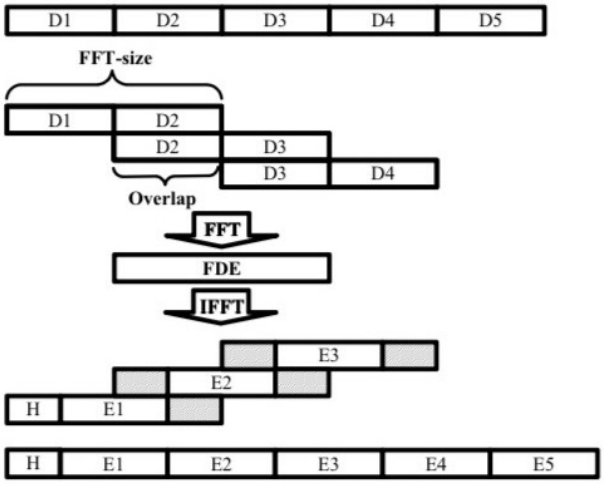


Figure 2: Schematic Representation of OLS method

3. **Apply IFFT:**

   Transform back to time domain:

$$y_k[n] = \text{IFFT}_N\{Y_k[\omega]\} \tag{18}$$

4. **Discard Overlap:**

   Keep only the valid samples $y_k[R]$ to $y_k[N-1]$ from each block.

5. **Concatenate Valid Portions:**

   Combine the valid portions from all blocks to form the complete output sequence.

3.2 Overlap-Add Zero-Padding Methods

3.2.1 One-Side Zero-Padding (OLA-OSZP)

In the OLA-OSZP method, non-overlapping input blocks of length L are zero-padded to length $N = L + M - 1$:

$$x_{\text{zp}}^{(k)}[n] = \begin{cases} x[kL+n], & 0 \le n < L \\ 0, & L \le n < N \end{cases} \tag{19}$$

The output blocks are combined using overlap-add:

$$y[n] = \sum_k y_k[n - kL] \tag{20}$$

where the overlapping portions are algebraically added.

3.3 Overlap-Add Both-Side Zero Padding (OLA-BSZP)

Overlap-Add Both-Side Zero Padding (OLA-BSZP) is an efficient block processing algorithm for fast convolution or frequency-domain filtering, applicable when processing long signals with FIR filters or for frequency-domain equalization (such as chromatic

dispersion compensation in optical systems). The method divides the input into non-overlapping blocks, pads both sides of each block with zeros (to prevent circular convolution artifacts), processes each block via the Discrete Fourier Transform (DFT, usually via FFT), and then stitches the output blocks by overlap-and-add.

Algorithm Steps

Let the input signal $x[n]$ of length L_x, and the FIR filter (impulse response) $h[n]$ of length L_h.

1. **Choose block length and zero padding:**
   - Let block length be N_b (number of original data samples per block).
   - Let the total zero-padding per block be N_p, where $N_p = N_l + N_r$.
   - N_l and N_r are the number of zeros padded to the left and right of each block, respectively. Often, $N_l = N_r = \lceil (L_h - 1)/2 \rceil$.
   - Set $N_{\text{FFT}} = N_b + N_p$ as the FFT size.
2. **Divide $x[n]$ into non-overlapping blocks** of N_b samples:
   $$x^{(m)}[n], \quad n = 0, 1, \ldots, N_b - 1; \; m = 0, 1, \ldots, M - 1.$$
   where $M = \lceil L_x / N_b \rceil$.
3. **Zero-pad each block on both sides:**
   $$x_{\text{zp}}^{(m)}[n] = \begin{cases} 0, & 0 \le n < N_l \\ x^{(m)}[n - N_l], & N_l \le n < N_l + N_b \\ 0, & N_l + N_b \le n < N_{\text{FFT}} \end{cases}$$
   for $n = 0, 1, \ldots, N_{\text{FFT}} - 1$.
4. **Zero-pad filter $h[n]$ to the same FFT size:**
   $$h_{\text{zp}}[n] = \begin{cases} h[n], & 0 \le n < L_h \\ 0, & L_h \le n < N_{\text{FFT}} \end{cases}$$
5. **Frequency-domain processing:**
   (a) Compute $X^{(m)}(k) = \text{FFT}\left[x_{\text{zp}}^{(m)}[n]\right]$
   (b) Compute $H(k) = \text{FFT}\left[h_{\text{zp}}[n]\right]$
   (c) Multiply: $Y^{(m)}(k) = X^{(m)}(k) \cdot H(k)$
   (d) Inverse FFT: $y_{\text{zp}}^{(m)}[n] = \text{IFFT}\left[Y^{(m)}(k)\right]$
6. **Extract valid output samples from each block:**
   $$y^{(m)}[n] = y_{\text{zp}}^{(m)}[n + N_l], \quad n = 0, 1, \ldots, N_b - 1$$
7. **Stitch (overlap and add) blocks to reconstruct final output sequence:** The $y^{(m)}[n]$ blocks are placed at relative position $n = mN_b$ in the final output. Any overlapping samples (from adjacent blocks, due to spreading by the filter) are added.

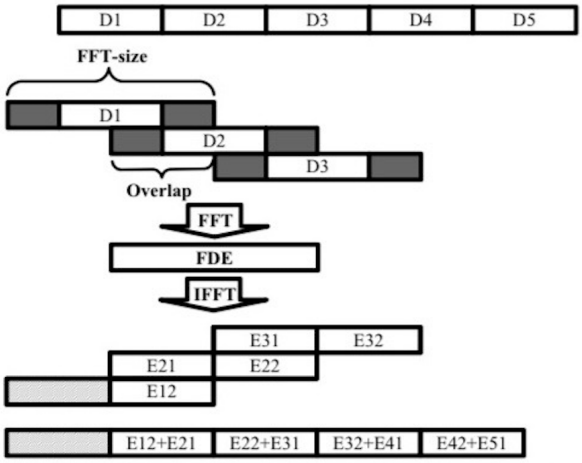


Figure 3: Schematic Representation of OLA-BSZP method

Advantages and Notes

- OLA-BSZP eliminates time-aliasing artifacts in block frequency-domain filtering and achieves perfect linear convolution if zero-padding is adequate (at least filter length $L_h - 1$ is split symmetrically on both sides).
- Compared to one-side zero-padding, the both-side variant limits boundary artifacts more efficiently and symmetrically.
- The required zero-padding/overlap depends on the filter's duration or the dispersive effect in, e.g., fiber optic systems [**?**].

3.4 Minimum Overlap Requirements

The theoretical minimum overlap is derived from pulse width broadening analysis:

$$N_P = \left\lceil \sqrt{1 + \left(\frac{4\pi cTDz}{\lambda^2 T^2} \right)^2} \right\rceil \tag{21}$$

where T is the sampling period and T_P is the broadened pulse duration.
The pulse width broadening is given by:

$$\frac{T_P}{T} = \sqrt{1 + \left(\frac{4\pi cTDz}{\lambda^2 T^2} \right)^2} \tag{22}$$

Practical Implementation:
Simulation results show that the practical overlap requirement N_S exceeds theoretical predictions by approximately 15%:

$$N_S \approx 1.15 \times N_P \tag{23}$$

3.5 Computational Complexity Analysis

The number of multiplications per symbol for FFT-based convolution is:

$$\frac{\text{Mul}}{\text{Sym}} = \frac{C \cdot N_{\text{FFT}} \log_2 N_{\text{FFT}}}{N_{\text{FFT}} - N_{\text{overlap}}} \tag{24}$$

where:

- $C = 1/2$ for radix-2 FFT
- $C = 3/8$ for radix-4 FFT

The optimum FFT size minimizing computational complexity satisfies:

$$\frac{d}{dN_{\text{FFT}}} \left(\frac{\text{Mul}}{\text{Sym}} \right) = 0 \tag{25}$$

3.6 Simulation Parameters

The system was simulated with the following parameters:

Table 1: System Parameters

Parameter	Value
Sampling frequency (F_s)	100 GHz
Dispersion parameter (D)	17 ps/(nm · km)
Wavelength (λ)	1550 nm
Fiber length (L)	1000 km
Pulse width (FWHM)	5 ps
FFT size (N_{FFT})	1024
Block size (K)	768

3.7 Mathematical Model

The chromatic dispersion is modeled by the frequency-domain transfer function:

$$H(\omega) = \exp\left(-i \frac{\beta_2 L \omega^2}{2} \right) \tag{26}$$

where $\beta_2 = -\frac{D\lambda^2}{2\pi c}$ is the group velocity dispersion parameter.
The compensation filter is its inverse:

$$H_{\text{comp}}(\omega) = \exp\left(i \frac{\beta_2 L \omega^2}{2} \right) \tag{27}$$

3.8 Results

Pulse Width Analysis

The pulse width evolution through the system:
The compensation achieved 99.2 % correlation with the original signal.

Table 2: Pulse Width Measurements

Signal	RMS Pulse Width
Original	150.14 ps
After dispersion	737.52 ps
After compensation	150.14 ps

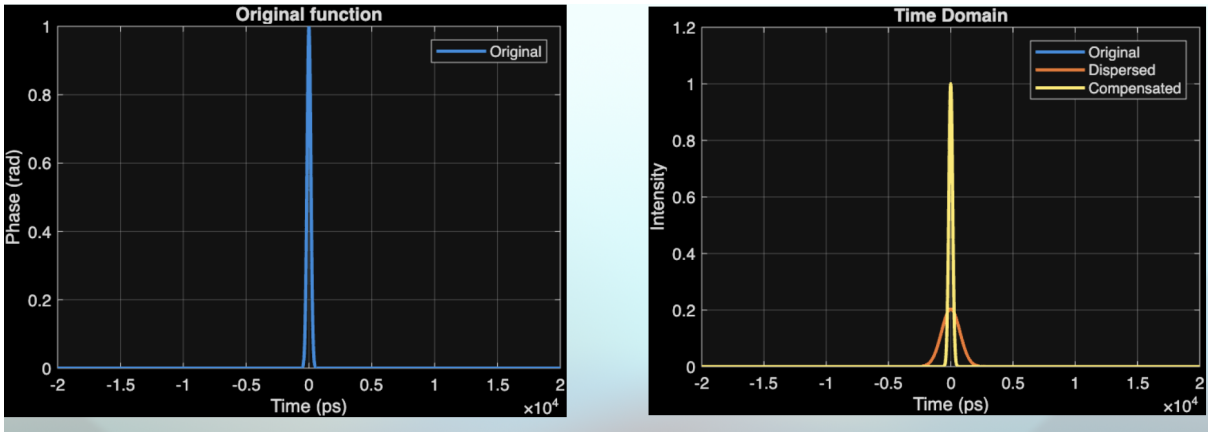


Figure 4: Application on Gaussian Pulse with attenuation

4 Pilot-Assisted Intradyne Reception for CV-QKD

4.1 Transmitter Setup

- A CW laser at $\lambda_T = 1550.12$ nm is pulse-carved to 250 MHz using a Mach-Zehnder modulator.
- Pulses are split into two orthogonal polarizations using a polarization-multiplexed I/Q modulator.
- The quantum branch is QPSK-modulated at 250 Mbaud; the pilot branch is modulated with a 1 GHz cosine using optical single-sideband modulation with suppressed carrier (oCS-SSB).
- The quantum signal is attenuated by 23 dB relative to the pilot for security and SNR optimization [1, ?].

4.2 Receiver Setup

- The receiver uses a free-running LO laser with linewidth < 20 kHz.
- A polarization-diversity 90° hybrid mixes the quantum and pilot signals with the LO and routes them to balanced detectors.
- Quantum data is detected using low-noise receivers; the pilot uses high-bandwidth receivers.
- Data is digitized at 20 GS/s for offline digital signal processing [1, ?].

4.3 Digital Signal Processing (DSP)

1. **Filtering:** Quantum data is lowpass-filtered at 250 MHz; pilot is bandpass-filtered at 4 MHz.
2. **Frequency Offset Estimation:** Phase drift between pilot and LO is estimated from the pilot's phase rotation.
3. **Phase Recovery:** QPSK data is raised to the fourth power to align constellation points, then back-rotated.
4. **Clock Synchronization:** Correlate with PRBS7 sequence.
5. **Parameter Estimation:** Calculate excess noise and SNR.

5 Experimental Results and Analysis

5.1 Excess Noise Calculation

The total quadrature variance at the receiver is:

$$V_B = TV_{\text{mod}} + N_0 + \xi$$

where T is channel transmittance, V_{mod} is modulation variance, $N_0 = 1$ SNU (shot noise), and ξ is excess noise.

For simultaneous I/Q measurement:

$$V'_B = \frac{T}{2}V_{\text{mod}} + N_0 + \frac{\xi}{2}$$

The conditional variance:

$$V'_{B|A} = 1 + \frac{\xi}{2}$$

Thus, excess noise is:

$$\xi = 2(V'_{B|A} - 1)$$

The SNR is:

$$SNR = \frac{TV_{\text{mod}}/2}{1 + \xi/2} = \frac{V'_B}{V'_{B|A}} - 1$$

[1]

5.2 CD Equalization Performance

Simulation shows that overlap-add methods (both OLA-OSZP and OLA-BSZP) achieve the same performance as the overlap-save method, provided the overlap (or zero-padding) is at least 15% greater than the theoretical minimum [2].

6 I Q Imbalance

Continuous Variable (CV) QKD protocol tackles the generation and distribution of a symmetric cryptographic key between two legitimate parties by using the employed telecommunication equipments. This approach reduces the cost effectiveness of the system, hardware complexity and increases the compatibility of QKD system with the available telecommunication equipments. One of the simplest and easier technique is to encode and transmit the bits for long distance is Quadrature Phase Shift Keying (QPSK) modulation format. For CV QKD, this technique belongs to the Discrete-Modulated (DM) category. The traditional communication system uses I(Inphase)-Q(Quadrature) modulators to create the QPSK modulation. Now, by the fundamental principle of QPSK modulation the I-component and Q-component should be always orthogonal to each other which can be represented by the following expressions;

$$x(t) = ACos\omega t + ASin\omega t \tag{28}$$

Here, the first term of the x(t) expression indicates the I-component term and second term indicates Q- component which are exactly 90° phase shifted to each other processing same amplitude throughout.

However, implementation flaws like misaligned I-, Q-, and phase port bias points, a poor coupler splitting ratio, a mismatched photodiode responsivity, and improper polarization controller adjustment can result in amplitude and phase imbalances that disrupt the orthogonality between the two received I and Q components and impair the QPSK system's performance. Furthermore, in real-world applications, phase diversity receivers are vulnerable to flaws in the optical 90° hybrid, which cause dc offsets as well as amplitude and phase errors in the signals that receive. In a communication system, these effects result in quadrature imbalance [ref]. Due to the mentioned nonlinearities, the QPSK modulators suffer from some phase and amplitude diversities which changes the balanced x(t) to

$$\begin{aligned} x_{imb.}(t) &= a_1 ACos(\omega t + \theta) + a_2 ASin\omega t & (29)\\ &= a_1 ACos\theta Cos\omega t + (a_2 - a_1 Sin\theta) Sin\omega t & (30)\\ &= A_{I,imb.} Cos\omega t + A_{Q,imb.} Sin\omega t & (31) \end{aligned}$$

Here, a_1, a_2 and θ are the gain imbalance of I-component, Q-component and phase diversity of the QPSK signal respectively. $A_{I,imb.}$ and $A_{Q,imb.}$ are the new amplitude of the QPSK signal which arises due to the imperfections of the hardware components. From the $x_{imb.}(t)$ expression it is pretty obvious that $A_{I,imb.} \neq A_{Q,imb.}$ and confirms the I-, Q- imbalance.

6.1 I Q Imbalance Compensation Procedure

Nowdays, use of digital signal processing (DSP) routine is a very trustworthy procedure to compensate the transmission impairments. In the context of compensation of I Q imbalance we used *Gram–Schmidt orthonormalization procedure* (GSOP) [5] at our DSP unit.

6.1.1 Description of *GSOP* :

Suppose $\{v_1, v_2, ..., v_n\}$ is a basis set for some vector space V with an inner product. By using GSOP we can produce an orthonormal basis set $\{w_1, w_2, ..., w_n\}$ for the same vector

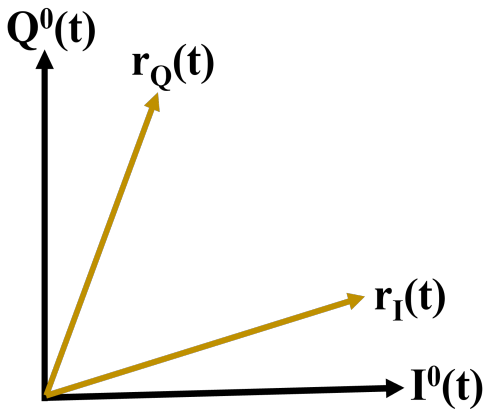


Figure 5: Representations of non-orthogonal received I ($r_I(t)$) and Q ($r_Q(t)$) signals w.r.t orthogonal basis function signals $I^0(t)$ and $Q^0(t)$.

space V. At the first step of this algorithm, $|w_1\rangle$ will be defined as the normalized version of $|v_1\rangle$ i.e. $|w_1\rangle = \frac{|v_1\rangle}{|||v_1\rangle||}$. Constructing orthonormal system$\{w_1,w_2,....,w_{k-1}\}$ whose span is exactly similar of $\{v_1,v_2,....,v_{k-1}\}$ vector space. To estimate the $|w_k\rangle$, we have to eliminate the projections onto the $|v_k\rangle$ from $|v_k\rangle$ and then normalize it. the final expression looks like;

$$w_k = \frac{|v_k\rangle - \sum_{j=1}^{k-1}\langle w_j|v_k\rangle|w_j\rangle}{|||v_k\rangle - \sum_{j=1}^{k-1}\langle w_j|v_k\rangle|w_j\rangle||}; \{k = 2, 3, ..., n\} \tag{32}$$

The new basis has the property for each k $\leq$ n and $span\{w_1,w_2,...w_k\} = span\{v_1,v_2,...v_n\}$.

6.1.2 Application of *GSOP* on I Q imbalanced signal :

The *GSOP* enables a pathway to transform a set of non-orthogonal samples to a set of orthogonal samples. As already mentioned due to multiple device imperfections, impairments, the I and Q components of the received signal becomes non-orthogonal to each other. With the application of the *GSOP* we will transform the received non-orthogonal $r_I(t)$ and $r_Q(t)$ component to a basis set of orthogonal functions $I^0(t)$ and $Q^0(t)$ [see Fig[5]].

At the first step of *GSOP*, we have to normalize the received $r_I(t)$ signal to define the $I^0(t)$ component i.e. $I^0(t) = \frac{r_I(t)}{\sqrt{P_I(t)}}$; where $\sqrt{P_I(t)}$ is the projection of $r_I(t)$ signal onto $I^0(t)$ and $P_I(t) = E\{r^2{}_I(t)\}$. Now to estimate the orthogonal Q signal i.e. $Q^0(t)$, we have to evaluate an intermediate function $Q'(t)$, this function will eliminate the projections of $I^0(t)$ signal from the $r_Q(t)$ signal, therefore the function can be described $Q'(t) = r_Q(t) - \rho I^0(t)$; where $\rho = E\{r_I(t).r_Q(t)\}$ is the correlation co-efficient. Finally to evaluate the orthogonal $Q^0(t)$ basis signal we have to normalize $Q'(t)$ i.e. $Q^0(t) = \frac{Q'(t)}{\sqrt{P_Q}}$; where $\sqrt{P_Q}$ is the projection of $Q'(t)$ onto $Q^0(t)$ and P_Q is defined by $E\{r^2{}_Q(t)\}$ ($E\{.\}$ denotes the average of ensemble operator). Therefore after the application of *GSOP* the new orthogonal signal is represented by $I^0(t) + jQ^0(t)$.

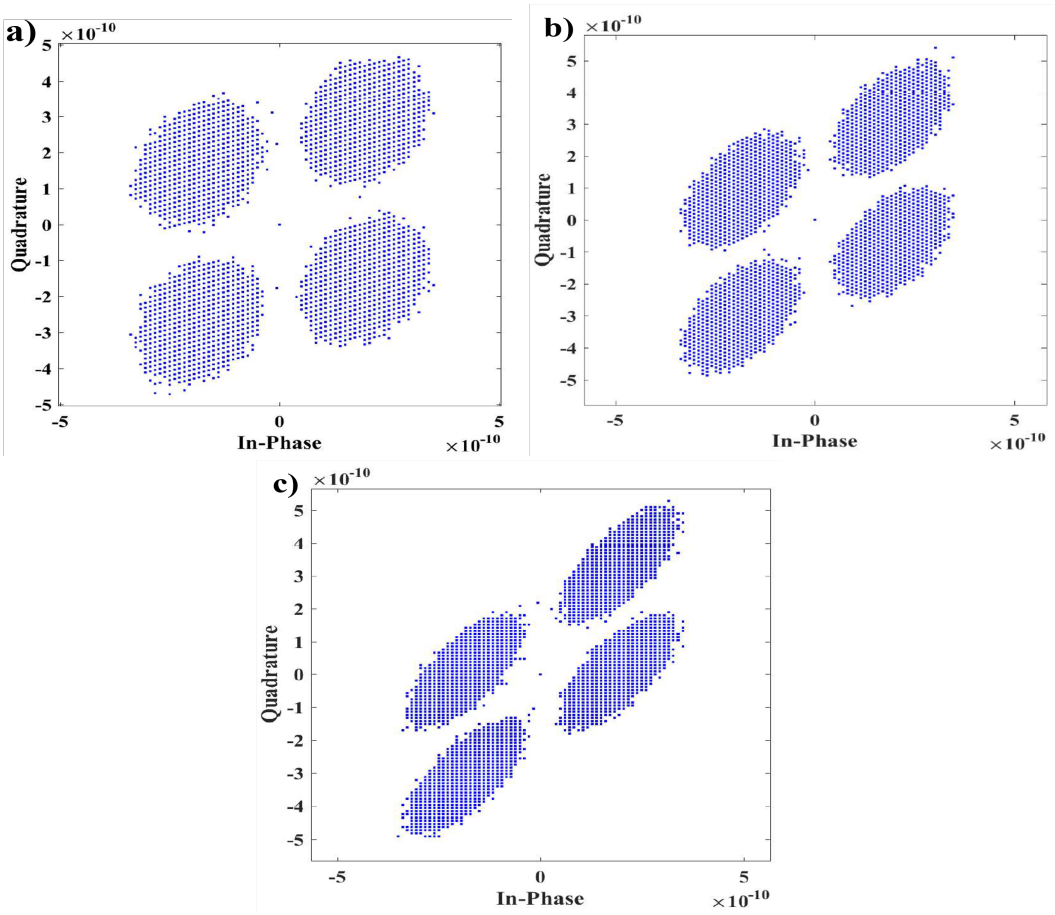


Figure 6: Representations of non-orthogonal received I and Q signals with (a) gain imbalance = 1.2, phase imbalance = 15 deg (b) gain imbalance = 1.2, phase imbalance = 30 deg (c) gain imbalance = 1.2, phase imbalance = 45 deg

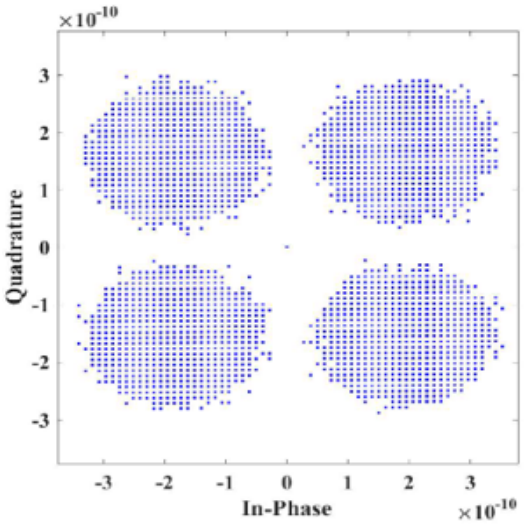


Figure 7: Representations of Orthogonal received I and Q signals by the application of modified GSOP.

6.2 Performance Evaluation of *GSOP* on I Q imbalanced signal

To evaluate the performance of the proposed algorithm, we intentionally introduce signal imbalances [see Fig.[6]] and subsequently apply the algorithm to the resulting non-orthogonal signals to assess its effectiveness. However, to preserve the quantum characteristics inherent in the communication system, we implement modifications to the conventional GSOP. Specifically, in the context of continuous-variable quantum key distribution (CV-QKD), a weak quantum signal is transmitted alongside a strong classical reference signal (pilot tone). It is crucial to ensure that the amplitude—and hence the variance—of the received quantum signal remains within the quantum regime. That is, the received variance V_B at Bob's side must remain less than the modulation variance V_A applied by Alice to preserve the quantum nature of the signal.

In the standard GSOP framework, normalization is typically applied to the received signal, which inadvertently amplifies its variance. Such amplification may push the signal beyond the quantum regime, thereby degrading its quantum properties. To address this, we propose a modified version of GSOP wherein normalization is deliberately omitted. By refraining from scaling the orthogonal components $I^0(t)$ and $Q^0(t)$, we avoid unnecessary amplification of the signal's amplitude and ensure that the quantum characteristics of the transmitted signal which are preserved throughout the orthogonalization process [see Fig.[7]].

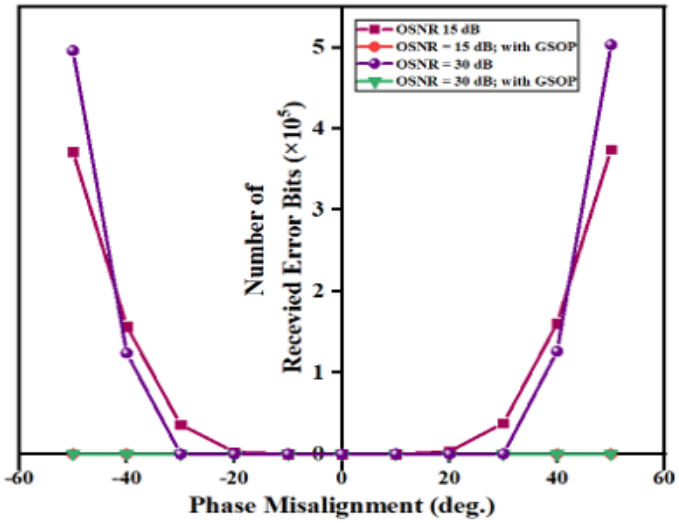


Figure 8: Error Estimation with and without modified GSOP for various Phase Misalignments

To further validate the robustness of the proposed protocol, we evaluate the performance of our modified GSOP algorithm under various practical impairment scenarios in Fig[8]. Specifically, simulations are conducted for two different optical signal-to-noise ratio (OSNR) values, both with and without the application of the modified GSOP, across a phase misalignment range of -50° to +50°, with a fixed gain imbalance factor of 2. The uncorrected, imbalanced signals (depicted in violet and purple) exhibit a significant increase in bit errors beyond approximately 30°–35° of phase misalignment at the receiver. However, when the modified GSOP is applied, the bit errorsis reduced to zero across the entire range for both OSNR levels (represented in red and green), demonstrating the algorithm's effectiveness in mitigating impairments arising from in-phase/quadrature (I/Q) imbalance.

These results confirm that the modified GSOP not only compensates for the detrimental effects of I/Q imbalance but also maintains performance even under low OSNR conditions—as low as 15 dB—highlighting the resilience and practical applicability of the proposed technique in realistic quantum communication environments.

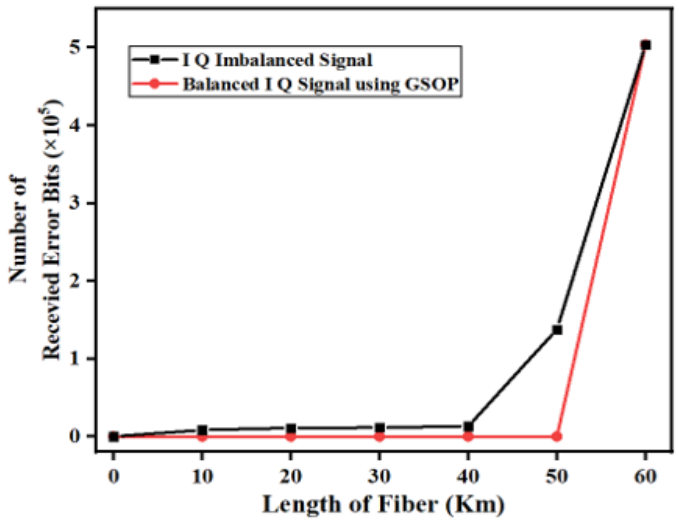


Figure 9: Error Estimation with and without modified GSOP for various lengths of fibre.

To assess the feasibility of the proposed approach for long-distance quantum communication under low OSNR conditions, we further simulate the system performance over varying fiber lengths at a fixed OSNR of 15 dB, this can be visualized in Fig[9]. The impairments are set to a fixed gain imbalance of 2 and a phase misalignment of 30°, and the results are compared with and without the application of the modified GSOP. Our simulation results reveal that, in the absence of GSOP, the imbalanced signal maintains an acceptable bit errors only up to a maximum transmission distance of approximately 40 km. In contrast, when the modified GSOP is employed, the maximum achievable distance increases to 55 km while still maintaining a low bit errors.

These findings demonstrate that the proposed GSOP-based compensation significantly extends the transmission distance under stringent conditions of low OSNR, severe I/Q imbalance and the algorithm demonstrates robust performance across a wide range of imbalance conditions, highlighting its adaptive nature and effectiveness in dynamically compensating for system impairments.. This highlights the potential of the algorithm to enhance the scalability and robustness of continuous-variable quantum key distribution (CV-QKD) systems in practical, noise-limited deployment scenarios.

7 RRC Filter Design and Integration for ISI Mitigation in CV-QKD

In continuous-variable quantum key distribution (CV-QKD), signal integrity during transmission over optical fiber is crucial for minimizing quantum bit error rate (QBER) and maximizing key generation rate. One of the major challenges affecting signal fidelity is inter-symbol interference (ISI), which arises due to pulse broadening and bandwidth

limitations in the transmission channel. To address ISI, a root-raised cosine (RRC) filter is employed for pulse shaping and matched filtering.

7.1 Inter-Symbol Interference (ISI)

ISI occurs when adjacent symbols in a digital communication system overlap in time, leading to distortion and degradation of signal detection. Mathematically, the received signal y(t) becomes a convolution of the transmitted pulse x(t) and the channel impulse response h(t):

$$y(t) = x(t) * h(t)$$

If the impulse response extends beyond the symbol interval, ISI is introduced. In the context of CV-QKD, where high-speed QPSK-modulated signals are transmitted over long distances, ISI becomes prominent due to fiber dispersion and limited bandwidth.

7.2 Need for Pulse Shaping

To mitigate ISI, pulse shaping is applied at the transmitter using the RRC filter. Pulse shaping ensures that each transmitted symbol occupies a controlled bandwidth and minimizes spectral leakage, while the time-domain pulse is shaped to reduce symbol overlap.

7.3 Matched Filtering

In digital communication systems like CV-QKD, pulse shaping is critical to control bandwidth and mitigate inter-symbol interference (ISI). The Raised Cosine (RC) filter is commonly used for this purpose due to its zero-ISI property. However, implementing a full RC filter at one end (transmitter or receiver) is computationally inefficient and hardware-intensive.

Instead, the RC filter is split into two square-root components—Root Raised Cosine (RRC) filters. One RRC filter is applied at the transmitter, and the other at the receiver. This configuration, known as matched filtering, ensures that the combined effect is equivalent to an RC filter, providing optimal signal recovery and minimum ISI.

$$h_{\mathrm{RC}}(t) = h_{\mathrm{RRC}}(t) * h_{\mathrm{RRC}}(t) \tag{33}$$

The matched filter maximizes the signal-to-noise ratio (SNR) at the sampling instant and is especially beneficial in quantum communication where signal power is weak and noise-sensitive. This structure also enables better control over bandwidth and simplifies hardware implementation.

7.4 Root-Raised Cosine (RRC) Filter

The RRC filter is derived as the square root of the raised cosine filter. The raised cosine (RC) filter is defined by its frequency response:

$$H_{RC}(f) = \begin{cases} 1, & |f| \leq \frac{1-\beta}{2T} \\ \frac{1}{2}\left[1 + \cos\left(\frac{\pi T}{\beta}\left(|f| - \frac{1-\beta}{2T}\right)\right)\right], & \frac{1-\beta}{2T} < |f| \leq \frac{1+\beta}{2T} \\ 0, & \text{otherwise} \end{cases}$$

Where β is the roll-off factor ($0 \leq \beta \leq 1$), and T is the symbol period. The RRC filter has the property:

$$H_{RC}(f) = |H_{RRC}(f)|^2$$

The impulse response of the RRC filter h_rrc(t) is given by:

$$h_{rrc}(t) = \frac{\sin(\pi t(1-\beta)/T) + 4\beta t/T \cdot \cos(\pi t(1+\beta)/T)}{\pi t/T\,[1-(4\beta t/T)^2]}$$

7.5 Filter Implementation

In our work, the RRC filter is implemented manually using the closed-form time-domain expression for h_rrc(t). The filter is truncated to a finite span (e.g., 12 symbols) and normalized to maintain power consistency. The filter is applied using time-domain convolution:

$$y(t) = x(t) * h_{rrc}(t)$$

To improve computational efficiency, FFT-based filtering may be used:

$$Y(f) = X(f) \cdot H_{rrc}(f)$$

7.6 Spectral Footprint Evaluation

To observe the spectral shaping effect, the power spectral density (PSD) of the signal is plotted before and after filtering using FFT:

$$S(f) = \left| \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j2\pi f n/N} \right|^2$$

This analysis confirms that the RRC filter confines the signal bandwidth and minimizes spectral leakage.

7.7 Spectrum Analysis Before and After RRC Filtering

Root Raised Cosine (RRC) filtering is critical in digital communication systems to control the spectral footprint and reduce Inter-Symbol Interference (ISI). This subsection presents the power spectral density of the QPSK signal at three key stages in the transmitter-receiver chain:

- Before any filtering (raw QPSK signal)
- After transmitter-side RRC filtering
- After receiver-side RRC matched filtering

The following figures demonstrate how RRC filtering sharpens the spectral boundaries and confines the signal bandwidth, ensuring efficient spectrum usage and better BER performance in Continuous Variable Quantum Key Distribution (CV-QKD) systems.

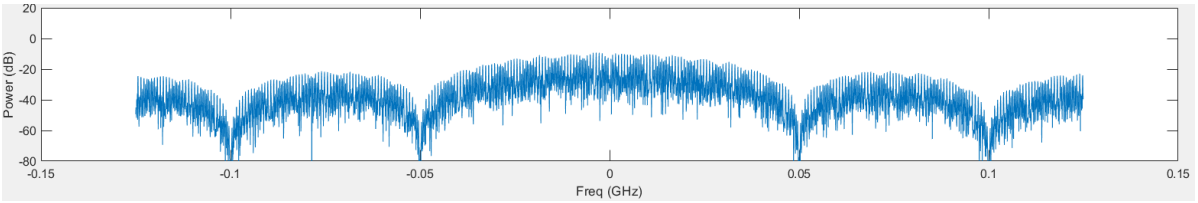


Figure 10: Spectrum of QPSK Signal Before Filtering

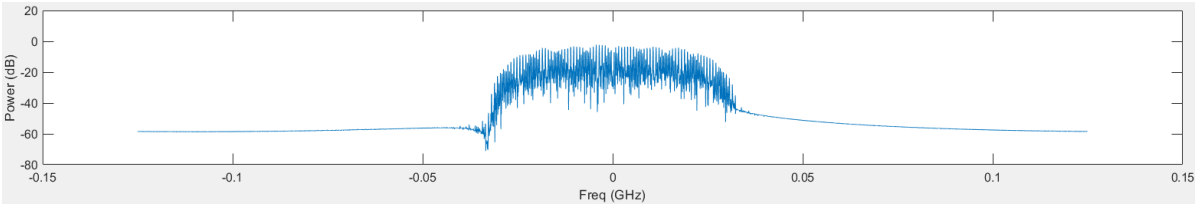


Figure 11: Spectrum After Transmitter-Side RRC Filtering

7.8 Observed Results and Performance Metrics

The spectrum plots at three key stages of the CV-QKD signal chain—before filtering, after transmitter-side filtering, and after receiver-side filtering—demonstrate the impact of Root Raised Cosine (RRC) filtering on bandwidth and signal quality.

- **Before RRC Filtering (Fig. 10):** The QPSK signal shows significant spectral spreading, with prominent side lobes and a flat passband extending over a wide frequency range. This is due to the sharp transitions in time domain from rectangular pulses, which cause high-frequency leakage. The unfiltered signal occupies a broader bandwidth, making it prone to inter-symbol interference (ISI) and spectral inefficiency.
- **After TX RRC Filtering (Fig. 11):** The transmitter-applied RRC filter smooths the time-domain waveform, which in turn suppresses side lobes and confines the bandwidth. The filtered signal exhibits reduced spectral width and roll-off at the edges, leading to better spectral efficiency and less interference to adjacent channels. This confirms the effectiveness of pulse shaping.
- **After RX RRC Filtering (Matched Filtering) (Fig. 12):** The receiver-applied matched RRC filter further sharpens the spectral confinement. The energy is concentrated within the main lobe, with significant attenuation in out-of-band components. This results in minimized ISI and maximized SNR, as expected from a matched filter. The final spectrum is compact and optimized for quantum detection.
- **Performance Metrics Observed:**

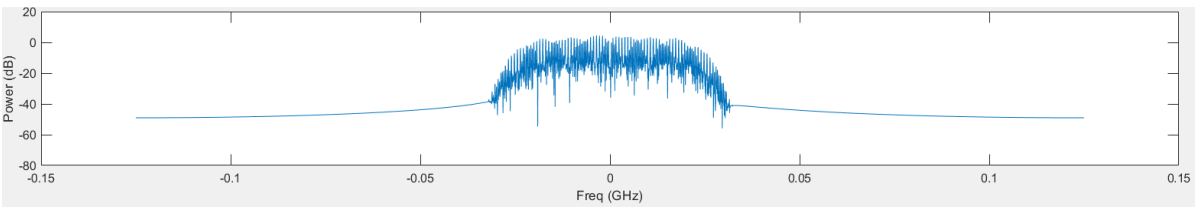


Figure 12: Spectrum After Receiver-Side RRC Matched Filtering

- **ISI Reduction:** Matched filtering significantly reduces inter-symbol interference, enabling clearer detection of quantum symbols.
- **SNR Improvement:** The RRC filter boosts SNR by focusing energy in the main lobe and suppressing out-of-band noise.
- **Bandwidth Control:** The spectral footprint is tightly controlled, allowing for high symbol rate operation without spectral leakage.

7.9 Future Work

- **Integration with Quantum Noise Models:** Future implementations should explore the integration of RRC filters with quantum noise models such as shot noise, squeezing noise, and excess noise. This would enable more accurate simulation of real-world CV-QKD systems, particularly in the presence of quantum uncertainties and practical imperfections.

8 Conclusion

Pilot-assisted CV-QKD using true local oscillator and frequency/polarization multiplexed pilot tones enables high symbol rates (250 Mbaud) and low excess noise (≤ 0.036 SNU over 40 km). Frequency-domain equalization using overlap-save and overlap-add methods provides efficient chromatic dispersion compensation with optimized computational complexity.Additionally, the integration of Root Raised Cosine (RRC) filtering within the QPSK modulation chain significantly enhanced the overall system performance. Pulse shaping and matched filtering through RRC suppressed inter-symbol interference (ISI), improved spectral efficiency, and supported better synchronization in high-speed quantum communications.

we have addressed the critical issue of in-phase/quadrature (I/Q) imbalance in CV-QKD systems by introducing a modified Gram–Schmidt Orthogonalization Procedure (GSOP) specifically designed to preserve the quantum nature of the signal. By avoiding amplitude normalization, the proposed approach ensures that the quantum signal remains within its required variance constraints, maintaining the integrity of quantum key distribution. Through extensive simulations, the algorithm has been shown to perform reliably under a variety of adverse conditions, including severe phase misalignments (up to 50°), pronounced gain imbalances, and low OSNR levels (as low as 15 dB). The results clearly demonstrate that the proposed technique not only achieves consistently low bit error rates under these impairments but also extends the secure transmission distance from 40 km to 60 km. These findings validate the robustness, efficiency, and adaptability of the modified GSOP, positioning it as a valuable solution for enhancing the practicality and scalability of long-distance CV-QKD systems in real-world, noise-limited environments. The combined framework of dispersion compensation and optimized filtering demonstrates a scalable approach for real-world CV-QKD systems requiring high data rates, low noise, and efficient DSP processing.

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